

Roll No.....

END SEMESTER EXAMINATION June-2025

Name of the Program: B. Tech (EE)

Semester: 6th

Name of the Course: High Voltage Engineering

Course Code: TEE 608

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Write down a short note on the thermal breakdown for solid material with suitable equations. What will the breakdown strength of air be for small gaps (1mm) and large gaps (20cm) under uniform field conditions and standard atmospheric conditions?	CO1
(b)	Define Townsend's first ionization coefficients. How is the condition for the breakdown obtained in a Townsend discharge?	
(c)	A Cockcroft-Walton type voltage multiplier has eight stages with capacitances, all equal to 0.05 μ F. The supply transformer secondary voltage is 125 kV at a frequency of 150 Hz. If the load current to be supplied is 5mA, find (a) the percentage ripple, (b) the regulation, and (c) the optimum number of stages for minimum regulation or voltage drop.	
Q2	(20 marks)	
(a)	A 100 kVA, 400V/250kV testing transformer has 8% leakage reactance and 2% resistance on 100 kVA base. A cable has to be tested at 500kV using the above transformer as a resonant transformer at 50 Hz. If the charging current of the cable at 500kV is 0.4A, find the series inductance required. Assume 2% resistance for the inductor to be used and the connecting leads. Neglect dielectric loss of the cable. What will be the input voltage to the transformer?	CO2
(b)	State and explain the Paschen's law. Drive the formula of Paschen's law with their curve diagram.	
(c)	Explain clearly the suspended particle mechanism of liquid breakdown with figures and suitable equations.	
Q3	(20 marks)	
(a)	What are the requirements of a sphere gap for measurement of high voltages? Discuss the disadvantages of the sphere gap for measurements.	CO3
(b)	Determine the breakdown voltage for air gaps of 2 mm and 15 mm lengths under uniform field and standard atmospheric conditions. Also, determine the voltage if the atmospheric pressure is 750 mm Hg and temperature 35°C.	
(c)	Explain with neat sketches Cockcroft-Walton voltage multiplier circuit. Explain clearly its operation when the circuit is unloaded.	

Q4	(20 marks)	CO4
(a)	Explain with neat diagram the principle of operation of an Electrostatic Voltmeter. Discuss its advantages and limitations for high voltage measurements.	
(b)	Draw a neat schematic diagram of Magneto Optic Method. An absolute electrostatic voltmeter has a movable circular plate 8 cms in diameter. If the distance between the plates during a measurement is 4 mm, determine the potential difference when the force of attraction is 0.2 gm wt.	
(c)	List out various tests to be carried out on the overhead line insulator and give a brief account of each test.	
Q5	(20 marks)	CO4, CO5
(a)	Draw a neat schematic diagram of the Expulsion type of lightning arrester with nomenclature. A 12-stage impulse generator has a $0.126 \mu\text{F}$ capacitor. The wave front and wave tail resistances are 800 ohms and 5000 ohms respectively. If the load capacitor is 1000 pF, find the front and tail times of the impulse wave produced.	
(b)	Explain the principle and operation of the surge absorber. What are ground rods, and what are their uses?	
(c)	What are the basic requirements of a surge diverter? Describe the construction, principle of operation and application of Rod gaps.	